

Strengthened Versions of a Uniform Property (S) Modulus Question

Follow-up note to the scalar counterexample packet

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Source and Question

This note follows up on the scalar counterexample packet for Open Problem (3) in:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The printed question asks whether $U(d, a)$ is always bounded above by a constant multiple of da . The scalar packet shows that, as printed, the answer is no: in the real one-dimensional space $\mathbb{R}$, for every $0 < d < 2$ the admissible pair in the infimum is forced to be the antipodal pair.

The purpose of this note is to record what happens after adding natural extra hypotheses such as infinite-dimensionality or the existence of small-distance pairs in the unit sphere.

Definitions

For a normed space X , Johnson and Kania define

$$U_X(x, y; a) = \sup_{\|z\| \leq a} | \|x + z\| - \|y + z\| | \quad (x, y \in S_X, a > 0),$$

and

$$U_X(d; a) = \inf_{\substack{x, y \in S_X \\ \|x - y\| \geq d}} U_X(x, y; a) \quad (0 < d < 2, a > 0).$$

Main Boundary Found

Theorem 1. *Let X be a non-superreflexive Banach space. Then*

$$U_X(d; a) \leq 2ad \quad (0 < d \leq 2, a > 0).$$

Consequently, the infinite-dimensional strengthened version of the printed question is true for all non-superreflexive Banach spaces.

Proof. Johnson and Kania prove, as Theorem C of the source paper, that every non-superreflexive Banach space satisfies

$$U_X(d; a) \leq \left(\frac{4a}{2 + d} \wedge 1 \right) d \quad (0 < d \leq 2, a > 0).$$

If $a \leq (2 + d)/4$, then

$$\left(\frac{4a}{2 + d} \right) d \leq 2ad,$$

because $2 + d \geq 2$. If $a \geq (2 + d)/4$, then $a \geq 1/2$, so the trivial estimate $U_X(d; a) \leq d$ gives $U_X(d; a) \leq d \leq 2ad$. $\square$

Corollary 1. *Any counterexample to the infinite-dimensional strengthened question must be superreflexive.*

The Flat-Face Mechanism

The source proof of Theorem C passes through almost-isometric ℓ_∞^2 geometry, equivalently a long line segment in the unit sphere after passing to an ultrapower. The elementary estimate behind that mechanism is the following.

Lemma 1 (Spherical segment estimate). *Suppose the unit sphere of X contains a line segment $[v, w]$ with $\|v - w\| = L > 0$. Then for every $0 < d \leq L$ and every $a > 0$,*

$$U_X(d; a) \leq \frac{4a}{L + d} d \leq \frac{4}{L} ad.$$

In particular, if $L = 2$, then $U_X(d; a) \leq 2ad$ for every $0 < d \leq 2$ and $a > 0$.

Proof. Parametrize the segment by

$$p(s) = (1 - s)v + sw, \quad 0 \leq s \leq 1.$$

For $0 < d \leq L$, put

$$x = p\left(\frac{1 - d/L}{2}\right), \quad y = p\left(\frac{1 + d/L}{2}\right).$$

Then $x, y \in S_X$ and $\|x - y\| = d$. Moreover

$$y = tx + (1 - t)w, \quad t = \frac{L - d}{L + d}.$$

Fix $z \in X$ with $\|z\| \leq a$. By interchanging x and y if necessary, assume $\|x + z\| \leq \|y + z\|$. Convexity gives

$$\begin{aligned} \|y + z\| &\leq t\|x + z\| + (1 - t)\|w + z\| \\ &\leq t\|x + z\| + (1 - t)(1 + a) \\ &= \|x + z\| + (1 - t)(1 + a - \|x + z\|). \end{aligned}$$

Since $\|x + z\| \geq 1 - a$, the final factor is at most $2a$. Also $1 - t = 2d/(L + d)$. Hence

$$\|y + z\| - \|x + z\| \leq \frac{4ad}{L + d}.$$

Taking the supremum over $\|z\| \leq a$, and then using this particular admissible pair in the infimum defining $U_X(d; a)$, proves the result. $\square$

Remark 1. *This lemma shows that merely having pairs at each small distance is not the decisive sufficient condition. What gives the estimate is a controlled family of pairs lying on a spherical segment.*

The Hilbert Case

Proposition 1. *If H is a real Hilbert space with $\dim H \geq 2$, then*

$$U_H(d; a) \leq 2ad \quad (0 < d < 2, \ a > 0).$$

Proof. Choose $x, y \in S_H$ with $\|x - y\| = d$. For every $z \in H$,

$$\|x + z\|^2 - \|y + z\|^2 = 2\langle x - y, z \rangle.$$

If $a \leq 1/2$ and $\|z\| \leq a$, then

$$\|x + z\| + \|y + z\| \geq 2(1 - a) \geq 1,$$

and therefore

$$\begin{aligned} \left| \|x + z\| - \|y + z\| \right| &= \frac{2|\langle x - y, z \rangle|}{\|x + z\| + \|y + z\|} \\ &\leq 2ad. \end{aligned}$$

Taking the supremum over such z gives $U_H(x, y; a) \leq 2ad$, hence $U_H(d; a) \leq 2ad$. If $a \geq 1/2$, then the trivial estimate

$$\left| \|x + z\| - \|y + z\| \right| \leq \|x - y\| = d$$

gives $U_H(d; a) \leq d \leq 2ad$. □

A Permanence Observation

Proposition 2. *Let $1 \leq p \leq \infty$, and suppose Y satisfies*

$$U_Y(d; a) \leq Cda \quad (0 < d < 2, \ a > 0).$$

Then every absolute sum $X = Y \oplus_p Z$ satisfies the same estimate.

Proof. It suffices to use admissible pairs inside the copy of Y . Let $x, y \in S_Y$, and view them as $(x, 0), (y, 0) \in S_X$. For a perturbation $(u, v) \in X$, with $1 \leq p < \infty$,

$$\|(x, 0) + (u, v)\|_p = (\|x + u\|^p + \|v\|^p)^{1/p}.$$

For fixed $r \geq 0$, the scalar map

$$s \mapsto (s^p + r^p)^{1/p}$$

is 1-Lipschitz on $[0, \infty)$. Hence

$$\begin{aligned} &\left| \|(x, 0) + (u, v)\|_p - \|(y, 0) + (u, v)\|_p \right| \\ &\leq \left| \|x + u\| - \|y + u\| \right|. \end{aligned}$$

The case $p = \infty$ is identical, using the 1-Lipschitz map $s \mapsto \max\{s, r\}$. Taking suprema over perturbations of norm at most a , and then infima over $x, y \in S_Y$, gives the claim. □

Remaining Target

This follow-up does not settle the connected-sphere strengthening. It also does not settle the infinite-dimensional strengthening in the superreflexive case. The reductions above show that a counterexample, if one exists, cannot be one-dimensional, non-superreflexive, Hilbert, a space with a diameter-two spherical segment, or an absolute ℓ_p -sum containing a good summand.

Thus any remaining infinite-dimensional counterexample must be superreflexive and must avoid both the flat-face mechanism and the Hilbert-type estimate. Numerical probes in the accompanying scratch script `lp2_modulus_probe.py` did not suggest an ℓ_p^2 counterexample: balanced pairs can be hostile for large p , but near-axis pairs become flat and appear to drive the infimum down.

Recommendation for the Original Packet

The original scalar packet should remain labeled as a literal counterexample to the printed question. Its limitation statement can be made sharper: the infinite-dimensional variant is already resolved positively for all non-superreflexive spaces by Johnson–Kania’s Theorem C, so only the superreflexive infinite-dimensional direction remains open from this follow-up.